Hedging the FX Carry Book with Futures: Funding-Currency Overlays and a Five-Class Comparison

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Summary

We test whether listed futures can hedge a 27-currency (G10 + EM) carry strategy (net Sharpe 0.466), following the assignment: hedge with funding-currency futures such as JPY — adjusting the hedge ratio, or hedging dynamically — and compare funding-currency, Treasury, S&P 500, DXY, and VIX futures. Every overlay is judged against a simple benchmark: cutting the book's exposure during FX-volatility stress, which alone raises the Sharpe to 0.571. **No futures overlay, static or dynamic, beats that benchmark.** Full analysis: `arjun/hedge_comparison.ipynb`.

1 Why not simply buy JPY?

The carry book is *short* JPY by design — it funds its high-yield longs by shorting the low-yielders, and that funding leg is the G10 book's single best earner. A standing long-JPY overlay is therefore just a resizing of a deliberately held, profitable position, and it pays the US–Japan rate differential every day it is on (−1.8%/yr at the hedged size). The only versions that could add value are conditional, so we test three designs of increasing sophistication:

- **Hedge-ratio adjustment** — a full sweep of static ratios h from −0.25 to 1, including the variance-minimising ratio $h^* = 0.52$;
- **Dynamic hedging** — the ratio re-estimated in real time (expanding and rolling one-year betas, applied with a one-day lag);
- **Stress-gating** — the hedge held only while a tradeable FX-volatility trigger is on (about one day in five).

The insurance is real: on the book's worst 5% of days the JPY overlay pays +34 bp/day, the best tail payoff of any instrument tested. But the carry premium eats the payout on every calm day. On honest pricing the results rank: de-risking (0.54) > unhedged (0.44) > stress-gated (0.39) > static h* (0.31) > dynamic (0.25–0.27). Dynamic hedging is *worst* because the JPY beta never changes sign — re-estimation adds noise, never timing.

2 The five futures classes compared

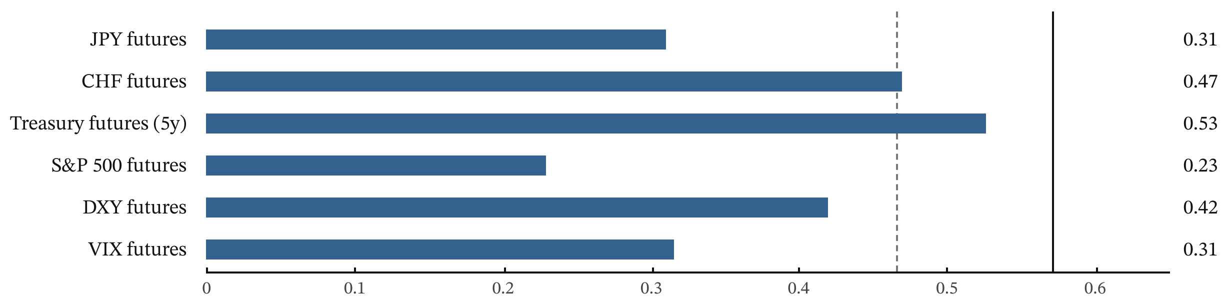

Figure 1. Net Sharpe of the book under each class's best static (variance-minimising) hedge, on artifact-free pricing. The dashed line marks the unhedged book (0.466); the solid line marks the de-risking benchmark (0.571). No hedge reaches the benchmark; most fall below doing nothing. Generic futures series overstate the funding-currency hedges by ~2%/yr of non-earnable roll gain; quoted figures use the forward-replica correction.

Table 1. Verdict by instrument class. "Static" is the variance-minimising fixed ratio; "dynamic" is the rolling-beta version.

Instrument class	Static	Dynamic	Assessment
JPY / CHF futures	NO	NO	Genuine crash insurance at full price; best tail payoff, but negative carry always wins
Treasury futures (2y–30y)	NO	NO	The only free hedge (right sign, no carry cost) — but too weak to matter (3–4% vol reduction)
S&P 500 futures	NO	NO	Wrong direction: the book moves *with* equities, so hedging sells the equity premium (−2.7%/yr)
DXY futures	NO	NO	Structurally blind: the carry sort is nearly dollar-neutral and DXY contains no EM currencies
VIX futures	NO	NO	Most reliable crash correlation, steepest cost (−33%/yr roll); viable only tiny and episodic

3 Conclusion

Everything that pays in a carry crash — long JPY, long volatility, short equities — is priced as insurance and charges for it on every calm day; everything free to hold is either too weak (Treasuries) or blind to the book's emerging-market risk exposure (DXY). The cheapest crash protection is therefore not a hedge but a rule: **reduce the book's own exposure when FX volatility spikes**, which raises the net Sharpe from 0.466 to 0.571 — more than any futures overlay tested, at essentially no cost.